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Controlled Formation of Nanostructures with Desired Geometries: Part 3. Dynamic Modeling and Simulation of Directed Self-Assembly of Nanoparticles through Adaptive Finite State Projection

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ABSTRACT: Deterministic dynamic modeling of self-assembled nanostructures, directed by external fields, through a master equation approach, leads to a set of differential equations of such large size that even the most efficient solution algorithms are overwhelmed. Thus, model reduction is a key necessity. This paper presents a methodological approach and specific algorithms, which generate time-varying, reduced-order models for the description of directed self-assembly of nanoparticles by external fields. The approach is based on finite state projection and is adaptive; that is, it generates reduced-order models that vary over time. The algorithm uses event-detection concepts to determine automatically, during simulation, suitable time points at which the projection space and thus the structure of the reduced-order model change, in such a way that the computational load remains low while the maximum simulation error, resulting from model reduction, is lower than a prescribed upper bound. Such a model reduction technique aligns well with a control strategy that modifies the strengths and locations of the external charges that direct the self-assembly, in order for the self-assembling system to achieve the desired geometry, while avoiding any kinetic traps. The paper also presents a series of case studies, which illustrate how the proposed method can be used to simulate effectively the directed self-assembly of an appreciable number of nanoparticles, avoid kinetic traps, and reach the desired geometry. These case studies will also illustrate several properties of the proposed methodology.

■ INTRODUCTION

Nanoscale systems provide the basis for technological advancements in various fields such as electronic devices, consumer products, multifunctional materials, diagnostic and therapeutic systems, molecular computing, and molecular factories.[1] Self-assembly[2,3] is a spontaneous or directed process, which offers fabrication routes toward nanoscale structures with unprecedented resolution. Significant progress has been made on self-assembly of *dense* nanoscale structures (i.e., structures with a very large number of particles, theoretically tending to infinity, extending over theoretically infinite domains), with *periodic* structural patterns, through the judicious design of building blocks to induce the desired short- and long-range orderings.[4] Such approaches fail to construct nanostructures with relatively small numbers of nanoparticles, within confined domains, and with nonperiodic structural features, as required by a number of applications, for example, nanoelectronics. Furthermore, even for systems with dense and periodic structures, fabrication techniques produce defects, i.e., failures in achieving the desired structure, usually as a consequence of kinetic traps. Consequently, the robust fabrication of nanostructured systems remains a major technological challenge, which becomes particularly acute when the desired structures are not dense, are confined in limited domains, and/or possess nonperiodic features.

Spontaneous self-assembly leads to free-energy structures only for systems with very large numbers of particles and unbounded domains. For systems with a relatively small numbers of particles, bounded domains, and nonperiodic desired structural features, external fields directing the self-assembling process are needed, such as shear,[5] magnetic,[6,7] or electric.[5,8−15] Electric fields are of particular interest because of their tunable strength and direction.[16−20] The key questions that needs to be answered are what are (a) the design parameters of the nanoparticles (shape, size, distribution of surface functionality), (b) the properties of the medium in which the self-assembly is taking place, and (c) the location and intensity of external fields, so that the nanoparticles assemble, with very high probability, to a structure that possesses the desired geometry?

In the present series of papers, we have attempted to answer this question by focusing first on item (c) of the question, that is, what are the locations and time-dependent intensities of the external controls, so that any initial arrangement of the nanoparticles will evolve to a final structure with desired geometry. In Part 1 of this series,[21] we addressed the static problem associated with the controlled formation of nanostructures. We were able to guarantee a statistically robust desired structure by solving the energy-gap maximization problem (EMP). We showed that we could reduce the phase space combinatorics and include only the neighboring competing states, while still guaranteeing a robust desired final structure. However, starting from an arbitrary distribution of particles in the complete phase space, the probability of reaching the desired nanostructure, under the influence of the static controls determined by the solution of the static problem in Part 1, is unacceptably low and

Special Issue: Scott Fogler Festschrift

Received: October 22, 2014
Revised: February 4, 2015
Accepted: February 10, 2015
Published: February 10, 2015

DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384

variable, since the system can be trapped in different metastable states, which are determined by the dynamic path of the self-assembly process. Thus, in Part 2[22] we developed a methodology that overcomes the above restriction and allows the dynamically self-assembled particles to reach the desired final geometry from any initial configuration with a prespecified high probability, that is, generate robust self-assembly paths. This methodology was based on the progressive reduction of the system phase space into subsets with progressively smaller numbers of locally allowable configurational states. In other words, it was based on a judicious progressive transition from ergodic to nonergodic subsystems. The subsets of allowable configurations in phase space that were modeled by a spatial multiresolution view of the desired structure (in terms of the number of particles) and the static optimal control were computed using the methodologies of Part 1. This approach produces a prescription for the optimal control, but not a truly optimal control policy, based on the dynamics of the evolving assembly process.

In the present, Part 3, and the upcoming, Part 4, of the series, we will use the prescriptions on optimal control, generated by the results of Part 2, and implement them in a true dynamic framework. Thus, Part 3 will focus on the dynamic modeling and simulation of the evolving configurations of nanoparticles, while Part 4 will generate and implement the optimal control strategies, using the dynamic description of the self-assembly process.

Lakerveld et al.[23] derived a dynamic model, involving master equations, to account explicitly for any kinetic traps in the course of a dynamic directed self-assembly process. The master equations form a probability-based continuous-time Markov process that captures the stochasticity of the system. Continuous-time models are essential for this system as discrete-time models would require very fine discretization to accurately model the stiffness in the system. Moreover, empirical discrete-time models require data from the full simulation of the system, which is difficult to obtain as the system size increases. The master equations-based model accounts for all potential configurations of the self-assembled nanoparticles and simulates the probabilities of observing the system in each specific configuration as a function of time, for given values of the control parameters. It also produces sensitivities of the dynamic profiles to the values of the control parameters, which play a very important role in computing optimal control policies. Although the favorable computational complexity of their method allows for the dynamic simulation of systems with fairly large numbers of configurations, it inevitably reaches its limits, as the number of configurations increases. Model-reduction techniques are needed to proceed.

An alternate model for a continuous-time Markov process is via the Langevin equations,[24] which are stochastic differential equations. There are various model reduction techniques used for Langevin equations, like dividing the equations into fast and slow dynamics,[25] and the method of adiabatic elimination.[26]

The reduction of models involving master equations has received considerable attention. The potential energy surface (PES) of the system can be simplified by aggregating configurations into shared basins of attraction, which in turn can form the basis for a reduced set of master equations, describing only the rates of rare events within a time scale of interest.[27,28] However, for optimal control problems, the separation of time scales is complicated since the features of the PES are a function of the unknown, as yet, optimal control policies. Other methods for model reduction include wavelets,[29−31] decomposition of ergodicity,[21,22] diffusion maps,[32,33] and finite state projection (FSP).[34−36] In the case of FSP methods, only those states which have a significant contribution to the dynamics of the system, known as the state projection space, are simulated. An FSP-based approach provides a bound on the maximum error that is being generated from the reduction of the state space, as a function of time. The advantage of using an FSP-based approach is that the mathematical structure of the model is preserved after model reduction, which permits exploitation of the favorable computational complexity, demonstrated by Lakerveld et al.[23] Furthermore, an FSP-based algorithm has two important properties that are essential for our purposes: (i) As we will show in this paper, it can be made to be adaptive, by changing the projection space in time; (ii) the logic of an adaptive FSP-based algorithm aligns well with the multiresolution approach developed in Part 2,[22] which uses a systematic decomposition of the systems ergodicity.

It should be noted though that dynamic models for self-assembly are known to be stiff, which would call for fast necessary changes in the projection space in only a small fraction of the total simulated time, when using FSP. Furthermore, to minimize total CPU time, one would like to anticipate when the maximum error, resulting from model reduction, grows too large, rather than taking a posteriori corrective action, through iterations backward in time. In this regard, the combination of an FSP-based algorithm and event detection[37] is of significant importance, in order to improve algorithms that simulate dynamic directed self-assembly processes.

Consequently, an adaptive FSP-based algorithm is well suited for the needs of our problem and the key questions that need to be answered are as follows: (a) When should the projection space be adapted, and what is the logic for its adaptation? (b) How should the projection space be adapted; that is, what configurations should exit the projection space, and what configurations should enter?

The answer to these two questions forms the essential scope and content of the present paper with the following elements:

1. The dynamic evolution of directed self-assembly of nanoparticles is modeled by master equations, describing the dynamic evolution of the probability of each configuration in projection space, that is, the space of selected configurations.
2. A variable-step ordinary differential equation (ODE) solver is used to integrate the master equations in time.
3. An event detection logic is employed to signal the need for adaptation of the projection space. This event-detection logic is integrated with the adaptive ODE solver.
4. The event detection logic employs an algorithm, which computes the upper bound on the error that is being generated, as a function of time, simultaneously with the probability of all configurations. Once this upper bound on the error crosses a certain prespecified threshold, the algorithm will precisely locate the point in time at which the threshold was crossed and, subsequently, adjust the projection space to reduce the rate at which the upper bound on the error is being generated so that the threshold is not crossed.

The resulting methodology modifies the projection space only when necessary, maintains control over the error in dynamic simulation, and allows the dynamic simulation of self-assembled nanoparticles at various spatial scales

DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384

(resolutions), as will be described in subsequent sections of this paper.

The paper is structured as follows: the next section contains all details of the methodological approach developed in this paper, that is, dynamic modeling of self-assembled nanoparticles through master equations; the adaptive FSP-based algorithm and its components; and the elements of the multiresolution approach. In the section after that, two case studies are presented. The first illustrates deployment of the adaptive FSP algorithm and demonstrates the benefits from its use in dynamically simulating complex self-assembled systems. The second case study demonstrates the multiresolution deployment of the adaptive FSP algorithm within the scope of a control strategy for the manufacturing of desired nanostructures. The theoretical generation and numerical implementation of optimal control strategies for self-assembled systems will be presented in the upcoming paper, Part 4, of this series.

■ APPROACH

A Dynamic Model for Directed Self-Assembly. The model for directed self-assembly is based on an Ising lattice model and the dynamics are described by a continuous-time Markov process. The probability of observing the particles in a particular configuration α as a function of time is given by a master equation:[38]

$$\frac{\mathrm{d}P_\alpha}{\mathrm{d}t}(t) = -\sum_{\beta\neq\alpha} \epsilon_{\beta\leftarrow\alpha}(t)P_\alpha(t) + \sum_{\beta\neq\alpha} \epsilon_{\alpha\leftarrow\beta}(t)P_\beta(t),$$
$$\forall\, t \in (t_0, t_f], \quad P_\alpha(t_0) = P_{\alpha,0} \tag{1}$$

where P_α is the probability of observing the system in configuration α and $\epsilon_{\beta\leftarrow\alpha}$ is a propensity factor, which denotes the rate of transition from configuration α to β. Equation 1 can be written in matrix form as

$$\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t}(t) = \mathbf{X}(t)\mathbf{P}(t), \quad \forall\, t \in (t_0, t_f], \quad \mathbf{P}(t_0) = \mathbf{P}_0 \tag{2}$$

where $\mathbf{P}(t)$ is a vector with probabilities for each configuration and $\mathbf{X}(t)$ is the matrix of propensity factors.

A propensity factor $\epsilon_{\beta\leftarrow\alpha}$ is obtained via transition-state-theory from the difference in potential energy between configuration α and β and normalizing it with an appropriate reciprocal time constant, ν. The energy of a configuration is constructed taking into account the pairwise interactions among the particles and the interactions between the particles and the external charges. The pairwise interactions among the particles are calculated from long-range repulsive electrostatic forces and short-range attractive forces such as van der Waals. The external electrostatic field is shaped by point charges on the grid, which are used as controls to direct self-assembly. The potential energy of a configuration α is given by

$$E_{0,\alpha}(t) = k_C(t)\sum_{i=1}^{V}\sum_{k=1}^{N_{\mathrm{EP}}} \frac{q_k(t)q_p}{|r_{i,k}|} z_{\alpha,i} + \sum_{i=1}^{V}\sum_{j\neq i}^{V} z_{\alpha,i}\left(\frac{k_C(t)q_p^2}{|r_{i,j}|} - \frac{a(t)}{|r_{i,j}|^6}\right) z_{\alpha,j} \tag{3}$$

where N_{EP} is the number of external charges, q_k is the strength of external charge k, q_p is the strength of the coulomb charge of each nanoparticle, k_C is a system-dependent parameter to compute the difference in potential energy that results from the electrostatic interactions, a is the parameter which lumps the effects of the close-range interactions, $z_{\alpha,i}$ is a binary variable which takes values 0 or 1 based on the absence or presence of a particle in cell i of the configuration α, $|r_{i,j}|$is the distance between nanoparticle i and j, and $|r_{i,k}|$is the distance from the center of cell i to point charge k. Note that the propensity factors will change with time, when time-dependent charges are utilized. The model is a phenomenological one, which can be extended to more complicated cases. The structure of the model and an algorithm to simulate large instances of this model are described in more detail by Lakerveld et al.[23] The model captures the inherent randomness in the movement of the nanoparticles, and can be extended to more complicated cases. The focus of this work has been to develop a computational methodology to solve this model.

Simulation of the system via FSP will result in a reduced model given by

$$\frac{\mathrm{d}P_\alpha}{\mathrm{d}t}(t) = -\sum_{\beta\in A(t):\beta\neq\alpha} \epsilon_{\beta\leftarrow\alpha}(t)P_\alpha(t) + \sum_{\beta\in A(t):\beta\neq\alpha} \epsilon_{\alpha\leftarrow\beta}(t)P_\beta(t),$$
$$\forall\, \alpha \in A(t), \quad P_\alpha(t_0) = P_{\alpha,0} \tag{4}$$

This reduced model will inevitably result in an error compared to a simulation of the full system, because the probability of observing unlikely configurations will be neglected. The upper bound on the total error as a result of model reduction is given by[34]

$$\varepsilon(t) = 1 - P_t(t) \tag{5}$$

where P_t is the sum of the probabilities of all the configurations in the projection space $A(t)$, described by

$$\frac{\mathrm{d}}{\mathrm{d}t}P_t = \sum_{\alpha\in A(t)} \frac{\mathrm{d}P_\alpha(t)}{\mathrm{d}t}, \quad \forall\, t \in (t_0, t_f],$$
$$P_t(t_0) = \sum_{\alpha\in A(t)} P_{\alpha,0} \tag{6}$$

The projection space $A(t)$ contains the likely configurations of the system. Note that this reduced model contains the connections that result in the out flux of probability from the projection space to the rest of the configuration space, but does not contain any connections that result in the influx of probability to the projection space. Therefore, detailed balance is not satisfied in the FSP model. There is no "steady state" in the reduced model and the error keeps increasing. However, once the system reaches a steady state, the increase in the upper bound for the error over a significant time scale will be small. In the next section, an adaptive algorithm will be presented to modify the projection space $A(t)$ during the simulation of directed self-assembly, whenever $\varepsilon(t)$ reaches an undesirable value. The aim is to include in the projection space the most relevant configurations (i.e., configurations of high probability), and thus approximate the dynamic behavior of the whole system with the lowest possible error, while avoiding the need to predefine suitable time-points at which the projection space should be modified; a task that is very difficult to complete a priori for directed self-assembly.

Algorithm: Adaptive Finite State Projection (AFSP). The key challenge for the design of an adaptive FSP-based algorithm is to define rules that identify the suitable time points during a simulation, at which points the structure of the projection space $A(t)$ should be changed. A trade-off exists between frequent changes in projection space to lower $\varepsilon(t)$ and avoiding

DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384

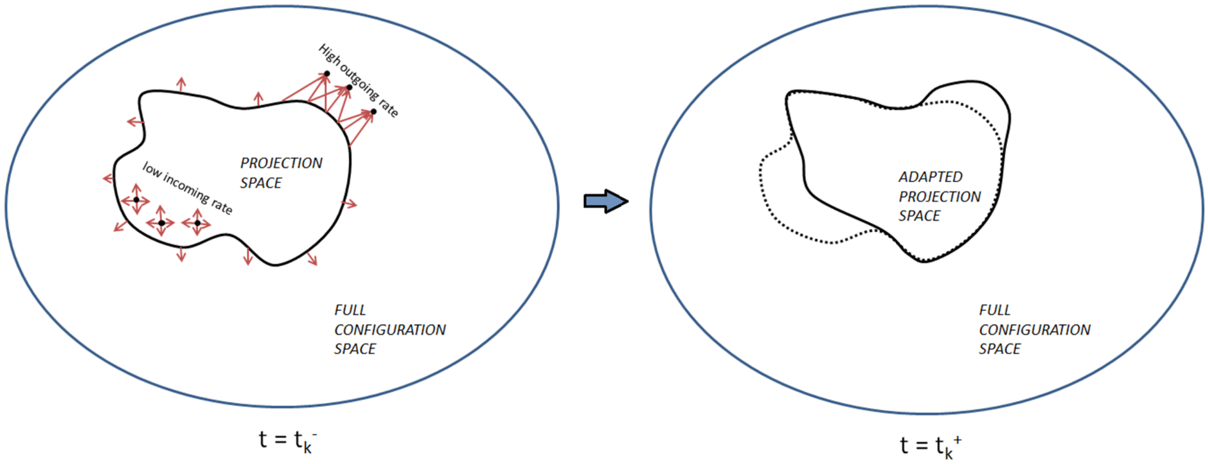

Figure 1. Illustration of the criteria for projection space adjustment in the proposed adaptive finite state projection method. Configurations outside the projection space receiving high transition rates will be added in the next time step. Similarly, configurations inside the projection space with low probability value and transition rates are removed at the projection space adjustment event.

a large computational load associated with each reformulation of eq 4, whenever the projection space has been changed. Therefore, time varying maximum and minimum allowed values for $\varepsilon(t)$ are introduced:

$$\varepsilon_{\min}(t) \leq \varepsilon(t) \leq \varepsilon_{\max}(t), \quad \varepsilon_{\min}(t) < \varepsilon_{\max}(t) \tag{7}$$

During a simulation, the upper bound on the error generated will be monitored and an algorithm will be designed such that eq 7 holds. A minimum value for the error generated allows for configurations, which can be truncated without sacrificing significant accuracy, to be removed from the projection space to reduce computational load. Furthermore, a maximum value for the upper bound on the error generated ensures that relevant configurations will be added to the projection space to achieve sufficient accuracy of the reduced model. The points in time at which the constraints described by eq 7 are crossed have to be identified and rules have to be defined to change the projection space at those times. The former requirement is addressed by implementation of an event detection scheme[37] and the latter requirement is addressed by implementation of the following rules:

1. Configuration α will be removed from the projection space $A(t)$ if and only if the following two conditions are met: (1) $P_\alpha(t) < P_{\text{trunc}}$ and, (2) $\mathrm{d}P_\alpha/\mathrm{d}t(t) < f_{\text{trunc}}$. These two conditions will ensure that only configurations with a low probability and low increase in probability will be removed.
2. Candidate configuration β will be added to the projection space $A(t)$ if and only if $\sum_{\alpha \in A(t)} \epsilon_{\beta \leftarrow \alpha}(t)\, P_\alpha(t) > f_{\text{add}}$, where α refers to all configurations that are part of the projection space and β refers to a single candidate configuration. This condition will add those configurations to the projection space that are likely to be important in the near future.

The resulting mechanism for addition and removal of configurations is schematically illustrated in Figure 1.

The application of the proposed algorithm is illustrated in Figure 2. The maximum and minimum allowed values for the upper bound on the error generated ($\varepsilon_{\text{MAX}}(t)$, $\varepsilon_{\text{MIN}}(t)$ respectively) are chosen to increase linearly with time so that an error budget at the final time is not exceeded. In this case, the initial probability distribution is assumed to be known and completely

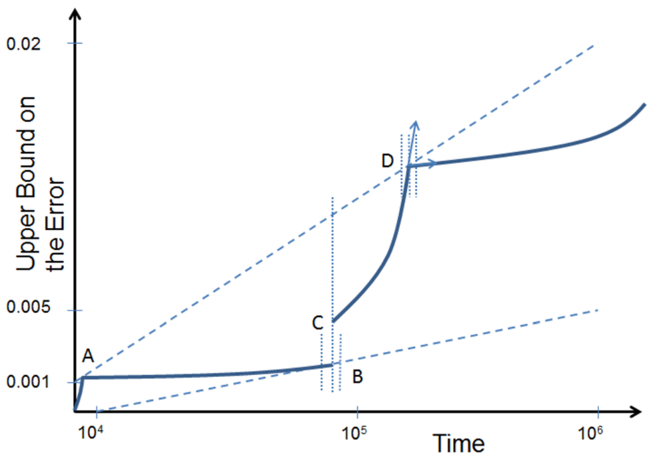

Figure 2. Example of simulation of directed self-assembly using an adaptive algorithm for finite state projection and event detection. The upper dashed line represents the maximum value allowed for the upper bound on the error resulting from model reduction ($\varepsilon_{\text{MAX}}(t)$) and the lower dashed line represents the minimum value allowed for the upper bound on the error ($\varepsilon_{\text{MIN}}(t)$). The solid line represents the upper bound on the error during simulation of a case study of directed self-assembly. The points A, B, and D are determined by event detection and indicate those points where the projection space is adjusted. The point C is the upper bound on the error after reinitialization of the model with the reduced projection space compared to point B.

included in the projection space. As time progresses, the upper bound on the error generated by the simulation increases until it surpasses the imposed maximum value $\varepsilon_{\text{MAX}}(t)$. Event detection[37] will locate the point A where $\varepsilon(t) = \varepsilon_{\text{MAX}}(t)$. Subsequently, configurations will be added to the projection to reduce the rate at which the upper bound on the error generated increases. After reinitialization of the system, the simulation of directed self-assembly proceeds until, in this case, the minimum allowed value is crossed. This indicates that configurations may be truncated to reduce computational load while still achieving an acceptable accuracy. Again, an event detection scheme precisely locates the crossing (point B in Figure 2). The truncation of configurations will result instantaneously in an increased upper bound on the error (point C) due to the nonzero value of the probabilities of the configurations removed from the projection space. Again, after

DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384

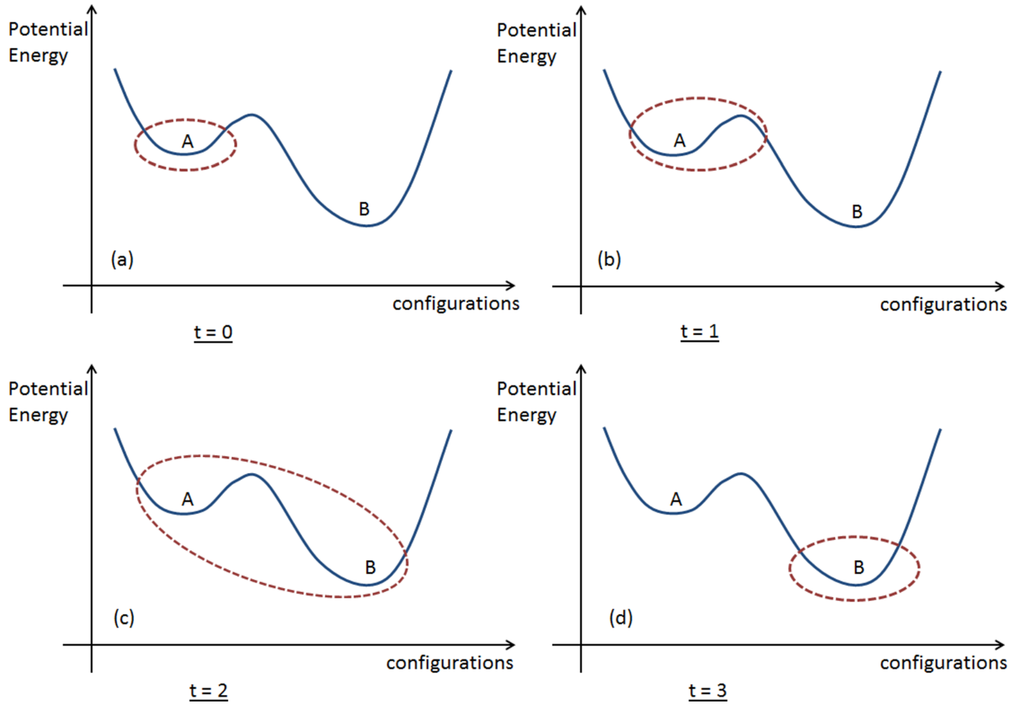

Figure 3. Schematic illustration of the proposed method, where the projection space (red dotted line) initially contains the configurations in (meta)stable state A. Eventually due to the transitions toward other (meta)stable states, the projection space expands to include the transition configurations and the stable configurations. However, as the probability evolves, the (meta)stable states lower their probability values and the projection space has only the configurations of B.

reinitialization, the simulation proceeds until the upper bound on the error hits the limits again (point D in the figure), and the process is repeated. The simulation provides the probability of observing self-assembled structures and a bound on the error resulting from model reduction, which is readily achieved without the need of repeating (part of the) the simulation or an arbitrary a priori discretization of the time interval.

The full algorithm proceeds as follows:

Step 0. Set $k = 1$ and set $t_k = 0$. Choose an initial projection space $A(0)$ that incorporates the initial probability distribution for the FSP.

Step 1. Solve eq 2 for one step with a variable-step ODE solver. Let that time interval be $(t_k, t_{k+1}]$.

Step 2. Compute $\varepsilon(t_{k+1}) = 1 - \sum_\alpha p_\alpha(t_{k+1})$.

Step 3. If $\varepsilon(t_{k+1}) > \varepsilon_{\mathrm{MAX}}(t_{k+1})$

(a) Find the coefficients of the polynomial fit of the data points computed by the ODE solver.

(b) Use Newton's method to locate the time t'_{k+1} of the crossing of the allowable limits on the error between t_k and t_{k+1} via interpolation.

(c) If $\sum_{\alpha \in A(t'_{k+1})} \epsilon_{\beta \leftarrow \alpha}(t'_{k+1})\, P_\alpha(t'_{k+1}) > f_{\mathrm{add}}$, then add configuration β. Repeat this procedure for all candidate configurations. A candidate configuration β is obtained by exploring all possible one-step transitions that lead to a new configuration outside the projection space.

(d) Augment $A(t'_{k+1})$ with the configurations added under step 3c.

(e) Calculate $(\mathrm{d}\varepsilon/\mathrm{d}t)(t'_{k+1}) = 1 - \sum_{\alpha \in A(t'_{k+1})} (\mathrm{d}p_\alpha/\mathrm{d}t)(t'_{k+1})$.

(f) If $(\mathrm{d}\varepsilon/\mathrm{d}t)(t'_{k+1}) > (\mathrm{d}\varepsilon_{\mathrm{MAX}}/\mathrm{d}t)(t'_{k+1})$, reduce f_{add} and go back to step 3c.

Step 4. If $\varepsilon(t_{k+1}) < \varepsilon_{\mathrm{MIN}}(t_{k+1})$

(a) Find the polynomial coefficients of the backward differentiation formula (BDF) method used by the ODE solver.

(b) Use Newton's method to locate the time t'_{k+1} of the crossing of the allowable limits on the error between t_k and t_{k+1} via interpolation.

(c) For all configurations in the projection space $A(t'_{k+1})$: truncate the configuration α if $p_\alpha(t'_{k+1}) < p_{\mathrm{trunc}}$ and $(\mathrm{d}p_\alpha/\mathrm{d}t)(t'_{k+1}) < f_{\mathrm{trunc}}$.

(d) If $\sum_{\alpha \in A(t'_{k+1})} \epsilon_{\beta \leftarrow \alpha}(t'_{k+1})\, P_\alpha(t'_{k+1}) > f_{\mathrm{add}}$, then add configuration β. Repeat this procedure for all candidate configurations. A candidate configuration β is obtained by exploring all possible one-step transitions that lead to a new configuration outside the projection space.

(e) Delete from $A(t'_{k+1})$ all configurations identified under step 3c and augment $A(t'_{k+1})$ with the configurations added under step 3d.

(f) Calculate $\varepsilon(t'_{k+1}) = 1 - \sum_{\alpha \in A(t'_{k+1})} p_\alpha(t'_{k+1})$.

(g) If ε is not within the error limits, go to step c after adjusting p_{trunc}, f_{trunc}, and f_{add}.

Step 5. Set $t_k = t'_{k+1}$ and go to step 1 if $t_k < t_f$.

This algorithm is different from the one by Park et al.[37] In this algorithm, the intermediate crossings of the allowable error limits after t_k are ignored if the upper bound on the error is within the error limits at t_{k+1}. This is because the algorithm aims to keep the upper bound on the error within the error budget only at the final time. Also, the final adjusted values of f_{add}, p_{trunc}, and f_{trunc} depend on the error limits and are determined automatically by the algorithm. The number of iterations required to include all the important configurations depends on the initial values of f_{add}, p_{trunc}, and f_{trunc}.

DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384

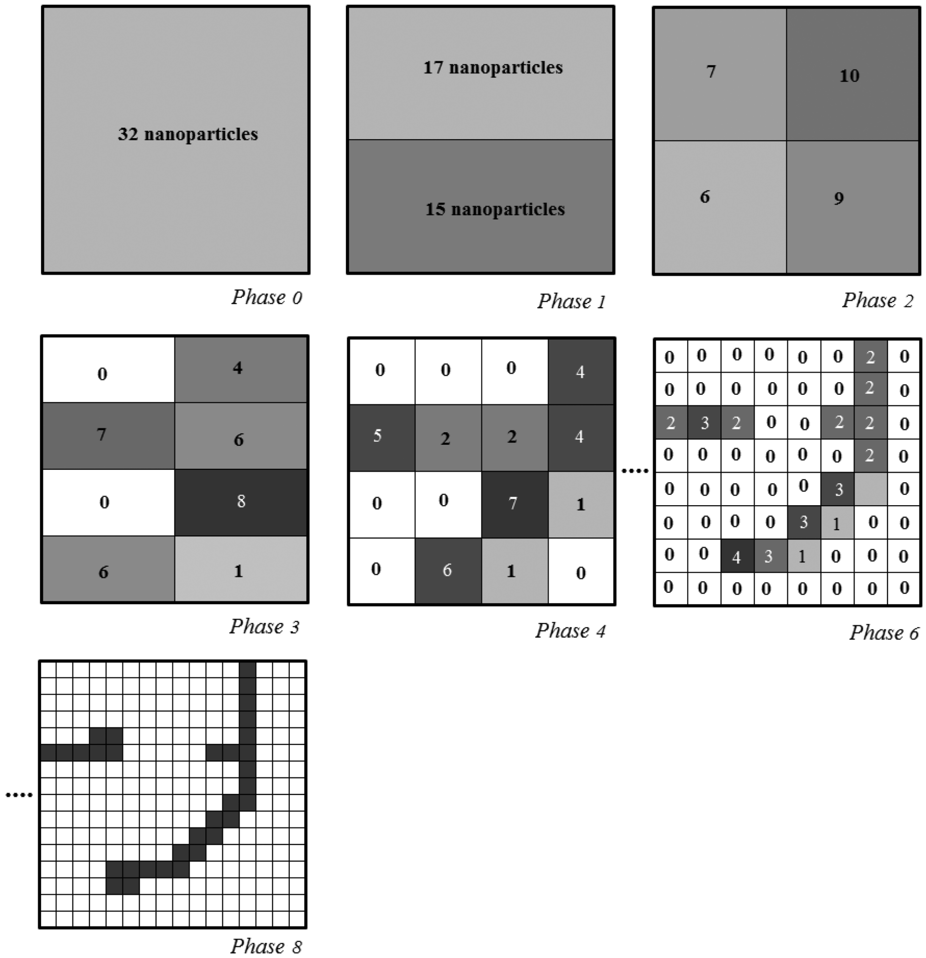

Figure 4. Illustration of the multiresolution approach used for self-assembly of nanoparticles. At every phase, the domain is divided into halves, and subsets of the nanoparticles are isolated in various parts of the domain. Self-assembly progressing through these multiresolution phases enables simulation of large domains.

An important point to note is that the inclusion of any configuration depends on the time constant of dynamic change in its probability value. A missing configuration that achieves a significant increase in its probability values within the time horizon of simulation will cause the upper bound on the error to increase rapidly, thus hitting the maximum allowed value, and this will result in the addition on that missing configuration. Hence, no important configurations can be missed in the proposed method.

Figure 3 illustrates schematically how the proposed algorithm responds to typical features of the potential energy surface over which the dynamics of self-assembly evolve. Suppose a system is initially trapped in a metastable state and the projection space of the reduced model involves configurations that have a common point of attraction (i.e., point A in Figure 3). The error as a result of model reduction in that case is caused by all transitions toward configurations that are not part of the projection space. Depending on the time interval of interest and the desired accuracy of the simulation, sufficient configurations may be added to include transition configurations between metastable state A to stable state B (i.e., Figure 3b,c). As the simulation proceeds in time, the probability of observing the system in a configuration with common attractor A will decrease and the algorithm will remove the configurations that enable the transition from A to B. Note that at this point the simulated system will not be ergodic anymore. In the next section, a case study will be presented in which the features of the potential energy surface are changed dynamically.

Multiresolution Approach. The multiresolution approach is used to tame the complexities arising during the dynamic simulation of self-assembly processes with large domains and many particles. In this approach, a desired nanostructure can be developed in phases, where at each phase the nanoparticles are arranged with progressively increasing resolution.

For example, in Figure 4, the self-assembly process starts with 32 nanoparticles which occupy any randomly selected initial configuration. In Phase 1, the external charges are positioned at appropriate points, and their intensities are selected in such a way as to maximize the probability of finding 17 and 15 nanoparticles in the top and bottom halves of the domain, respectively. Once this distribution has been achieved, transitions between the domains are restricted by strong repulsive external charges along their boundary. Subsequently, each of the domains of Phase 1 is treated independently in Phase 2. Each domain of Phase 2 is treated the same way as Phase 1 itself and further divided in Phase 3. This process continues until the final structure is reached. This is an illustrative example and the method can be extended to a system of any size.

There is no unique strategy for positioning the external charges and selecting their strengths in order to achieve the progressive multiresolution distribution of the particles in the domain, as shown in Figure 4. In an accompanying paper, Part 4 of this series, we will develop an optimal control policy, which achieves the multiresolution objectives, by minimizing

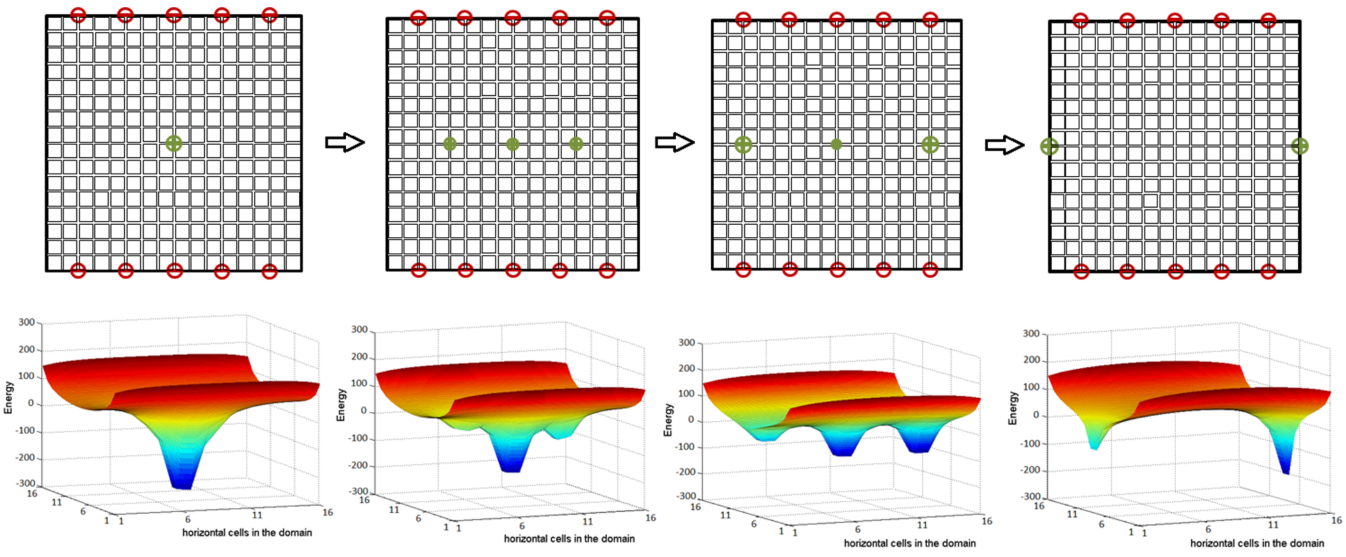

Figure 5. Illustration of the various steps of the proposed strategy for self-assembly in a particular step of the multiresolution approach. Each arrangement of external charges corresponds to an energy landscape, which gives an insight into the path that the nanoparticles will follow. The first step, shown in panel a, has a strong attractive charge in the center that attracts the nanoparticles toward it. In the subsequent steps, shown in panels b, c, and d, two additional charges are introduced, on either side of the charge in the center, that slowly increases in strength, while the strength of the center decreases. The net result is a progressive weakening of the center well and strengthening of two new wells, which move toward the edges of the domain, pulling along the corresponding number of particles, thus creating two distinct and weakly interacting sets of particles.

the required total time. For the dynamic simulation purposes of the present manuscript, to achieve the desired number of nanoparticles in each half, the locations of the external charges need to be suitably chosen and changed with time in such a way that the system does not get trapped in any kinetic traps. This will be demonstrated in the next section.

Designing the Energy Landscape for the Multiresolution Decomposition of the State Space. During the self-assembly process, the nanoparticles can be prevented from getting stuck in kinetic traps by suitably shaping the energy landscape of the system. Forming large energy wells in the energy landscape close to the desired location of the nanoparticles causes the nanoparticles to move toward these wells. These nanoparticles can then be moved to different locations of the domain by slowly moving the energy wells. Moving these wells slowly with time forms the basis of the proposed strategy. The energy wells are moved by progressively modifying the locations and strengths of the external charges and hence attract the nanoparticles along with them. This strategy is applied in combination with the AFSP method.

The dynamic simulation of the system, to achieve the first phase of the multiresolution approach, is started from an initial state where all the particles are near the center of the grid. This arrangement of nanoparticles is quite simple to achieve in practice—by placing a large attractive external charge in the center of the grid, as shown in Figure 5a. Once the probability of observing nanoparticles near the center of the grid is high, the arrangement of external charges is slightly modified. Two more external charges are added on either side of the charge in the center (left and right side in this case). These two external charges have a small attractive strength and the strength of the charge in the center is reduced by a small amount. This step creates two small energy wells on either side of the large energy well at the center (Figure 5b), and hence the probability of the desired separation of the nanoparticles will increase (for example, 17 particles above and 15 particle below in Phase 1 of Figure 4). After simulating the system for a fixed time, the two additional attractive external charges will be moved further away from the center, on either side of the charge in the center, with their strengths increased while the strength of the charge in the center is further reduced (Figure 5c). The simulation continues with this setup for a fixed time. This step causes the two energy wells on either side of the center well to deepen, with the energy well in the center becoming less deep, further increasing the probability of separation of the nanoparticles. This process will be continued until the additional external charges reach the edge of the grid, at which point, the arrangement of the external charges will be that given by the static solution (minimum tiling approach) proposed by Solis et al. in Part 1 of this series.[21] This solution is computed for a sample configuration of nanoparticles with the desired separation between them. The implementation of the static solution toward the end of the strategy will ensure that the probability of the desired separation of nanoparticles is quite high. The strengths of the external charges at the center will eventually go to zero or become negative (Figure 5d).

The above strategy of changing the locations and strengths of external charges is used at all phases of the multiresolution approach. Charges located at the boundary may be used to enhance or inhibit movement of particles to the boundary of the domain. As the multiresolution strategy decomposes the state space into nonergodic subsets of configurations, the effectiveness of the AFSP-based algorithm, described in the section on the algorithm, increases significantly and makes feasible the dynamic simulation of guided self-assembly processes with large-sized domains and large numbers of nanoparticles.

In the system used here for illustrative purposes, there are no energy wells in the energy landscape other than the ones specifically formed by the attractive external charges. This occurs because the external charges used have high strengths compared to the strengths of interactions between particles. Since kinetic trapping occurs when the number of nanoparticles

DOI: 10.1021/ie504182w
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trapped in a particular energy well is different than that desired, this can be avoided by suitably choosing the strengths of the external charges and monitoring the probability of the desired separation of the nanoparticles with time. During dynamic simulation, if the system is in a kinetic trap the rate of increase of the desired probability is very low, and the configurations in the projection space do not possess the desired final geometries. In such cases, the locations and intensities of the external charges must be changed in order to generate favorable energy landscapes that allow further evolution of the system toward the desired state. How to achieve this goal in an optimal manner, through the expanded capabilities of AFSP for the dynamic simulation of large systems, will be shown in the upcoming paper, Part 4 of this series.

■ CASE STUDIES

Dynamic Simulation through Adaptive Finite State Projection. The system under study consists of a 2D square domain with 36 lattice cells and 6 negatively charged nanoparticles. Self-assembly is driven by interactions between particles and interactions of the particles with an external electrostatic field shaped by various point charges. The total number of potential configurations for this system is 1 947 792. This total number of potential configurations is calculated by considering all possible ways in which six nanoparticles can be distributed on 36 lattice cells. The aim of the proposed algorithm is to simulate only a portion of those 1 947 792 configurations while maintaining sufficient accuracy in the dynamic evolution of the self-assembled system. The point charges are modified over time in such a way that the nanoparticles are directed from the top-right corner of the domain and eventually self-assemble with high probability into a structure with the nanoparticles in the bottom-left corner of the domain. During this transition, the majority of possible configurations will be kinetically restricted. The AFSP algorithm for simulation of this process will automatically take advantage of these kinetic restrictions, by modifying the projection space in time during the simulation.

The conditions for the simulation of the case study are given in Table 1. Note that a high value of (q_k/q_p) was used, which ensures that the electrostatic interactions with the external field are dominant.

Table 1. Parameter Values for the Simulation in the Case Study. N_t is the Number of Configurations in the Projection Space

parameter	value
strength of external charges (q_k/q_p)	+50 or −50
$k_C q_p^2$ [kcal mol^{-1}nm]	1
$k_B T$ [kcal mol^{-1}]	0.7
a [kcal mol^{-1} nm^6]	10^5
f_{add} or f_{trunc} [ν]	$P_t(t)/N_t(t) \times 10^{-3}$
p_{trunc}	$P_t(t)/N_t(t) \times 10^{-3}$

The parameter values of f_{add}, p_{trunc}, and f_{trunc} are provided in Table 1 as well. For each iteration in Steps 3 and 4 of the algorithm (see the section on the algorithm of AFSP), the values of f_{add}, p_{trunc}, and f_{trunc} were reduced or increased by 50%, as required by the algorithm. Reasonable initial estimates were chosen for these parameters.

The initial probability distribution of the studied case, for illustrative purposes, consists of six configurations with equal probability, while the probability of observing the system in any other configuration is initially zero. The expected number of particles per lattice cell corresponding to the initial distribution is illustrated in Figure 6a. The gray scale to the right of the physical domain indicates the expected number of particles that will occupy a cell in the domain. This number varies from 0 to 1, where 0 (white) indicates a particle will not occupy that grid cell and 1 (black) indicates a particle will occupy that grid cell with 100% probability. The expected number of particles in each cell is calculated by the sum of the probabilities of all the configurations that have a particle in that cell. Furthermore, Figure 6a illustrates the initial arrangement of the external charges. Note that this arrangement of the external charges will direct the nanoparticles to the bottom-right part of the domain as illustrated in Figure 6b. At $t = 1.0 \times 10^5\ \nu^{-1}$, the strengths of the external charges are changed instantaneously as illustrated in Figure 6c, which in turn will direct the nanoparticles toward the bottom-left corner of the domain. At the end of the simulation, a steady state is approached in which self-assembled structures with nanoparticles in the bottom-left corner of the domain have a high probability (Figure 6d).

The dynamics of the directed self-assembly process, as described above, clearly indicate the need to change the projection space during the simulation. Suitable time points to change the projection space are informed by the upper bound on the error resulting from model reduction, as explained in the section on the algorithm of AFSP. Figure 7 illustrates the upper bound on the error as a function of time, including the chosen maximum and minimum values allowed. The initial projection space contains only those six configurations for which $P_\alpha(t_0) > 0$ holds. The upper bound on the error initially increases rapidly and hits the maximum value allowed after a short time interval. Event detection locates the point in time where the upper bound on the error hits the maximum value and, subsequently, additional configurations are added after which the simulation proceeds. Around $t = 0.9 \times 10^5\ \nu^{-1}$ the minimum allowed value is crossed, which indicates that configurations may be removed from the projection space. After the update of the projection space, only a small number of (500) configurations are left, which are configurations with nanoparticles in the bottom-right part of the domain (Figure 6c). The simulation is continued until time $t = 1.0 \times 10^5\ \nu^{-1}$. At this point, the external charges are changed instantaneously to direct the nanoparticles to the bottom-left part of the domain. The modified external field causes the upper bound on the error to increase rapidly with the prevalent projection space and the maximum value is crossed rapidly (Figure 7b). Again, event detection locates the point of intersection and the projection space is modified subsequently to add new configurations based on the analysis of the propensity rates as described in the section on the AFSP algorithm. For the studied case, the sequence of event detection and expansion of the projection space is repeated until eventually the added configurations reduce significantly the rate at which the upper bound on the error increases. This can be seen in the magnification of a small region of Figure 7b, where A, B, ..., F show the time points when the upper bound on the error crosses the maximum value repeatedly. Figure 7a shows that simulation of the reduced model is indeed a close approximation of a simulation of the full system with all possible configurations and transitions.

In summary, the presented case study demonstrates the effectiveness of the proposed algorithm to simulate directed self-assembly by using an adaptive version of FSP for model reduction with event detection. Automated identification of a changing and limited number of relevant configurations and

DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384

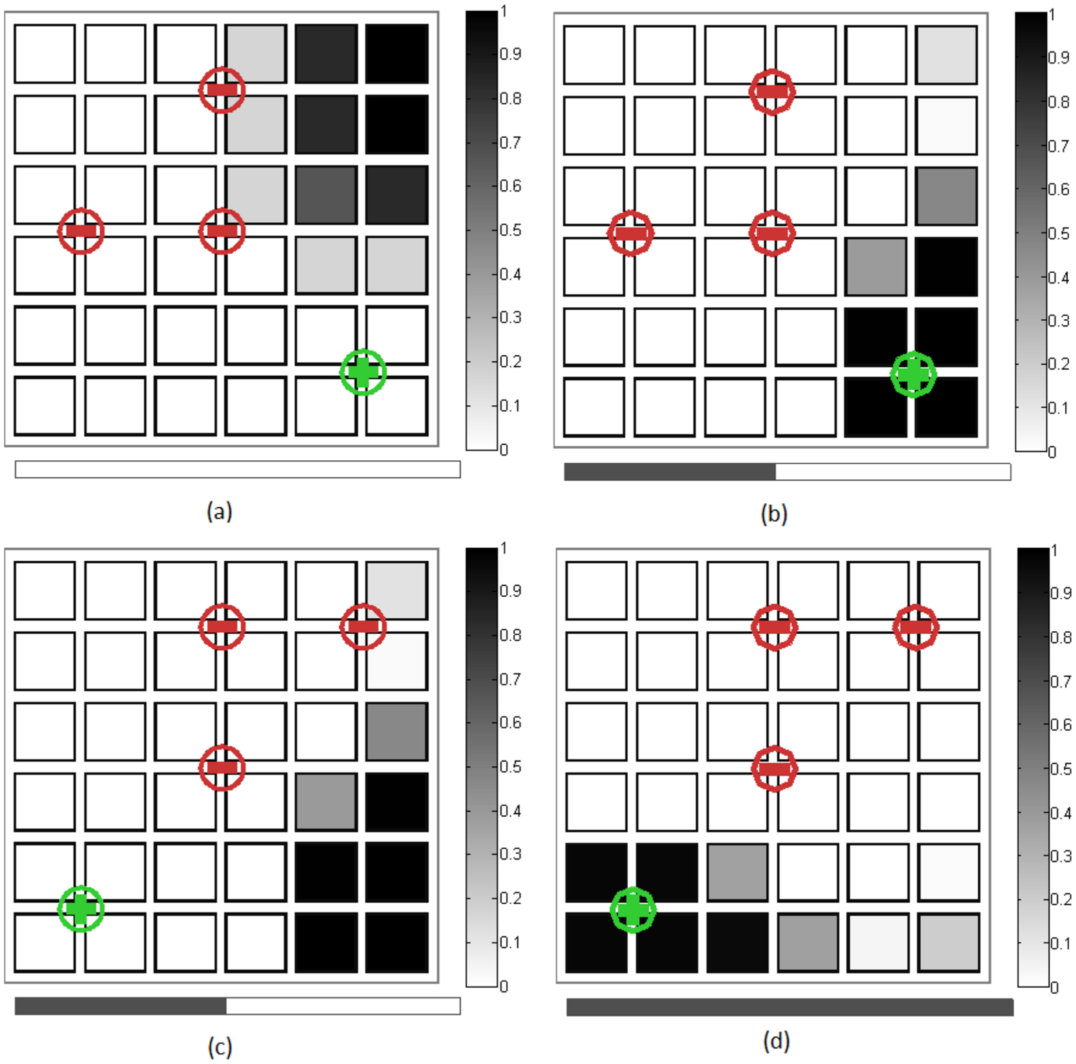

Figure 6. Snapshot of the dynamic evolution of self-assembly in the simulation at various times. The rectangle at the bottom of each grid is the progress bar which indicates the simulation time. The rectangle at the right of the grids shows the scale measuring the expected number of nanoparticles in a particular cell. Grid a shows the initial probability distribution in each grid cell along with the external charges used. After $t = 2.8 \times 10^6\ \nu^{-1}$, the probability distribution is as shown in grid b. At that instant, the external charges are changed as shown in grid c. Grid d shows the final probability distribution of the nanoparticles at $t = 6.1 \times 10^6\ \nu^{-1}$.

transitions (approximately 1% compared to the full system) is sufficient to obtain a desired accuracy on the maximum error resulting from model reduction. In the presented case, the time scales of desired changes in the state projection vary significantly, which does not pose restrictions since the approach does not require a priori information on desired changes of the state projection. It is expected that tuning of the parameters that quantitatively determine which configurations to add or truncate (i.e., f_{add}, p_{trunc}, f_{trunc}) can further enhance the performance of the algorithm for the studied case. Finally, it is expected that the proposed algorithm is particularly effective for systems in which the potential energy surface is strongly rugged, which is typical for directed self-assembly. Such systems can be much more complicated compared to the illustrated case; however, the automated nature of the algorithm allows such systems to be approached as well.

Control Strategy for Self-Assembly. This case study consists of a 2D square domain with 64 lattice cells and 8 negatively charged nanoparticles. As in the previous case, self-assembly is driven by interactions between particles and interactions of the particles with an external electrostatic field shaped by the various point charges. The total number of configurations for this system is 4 426 165 368, and it is not practical to simulate the full system with the algorithm of Lakerveld et al.[23] The aim of the proposed algorithm is to move the point charges and vary their strengths with time to facilitate self-assembly to a structure with desired geometry, while avoiding kinetic traps. Only a fraction of the 4 426 165 368 configurations need to be simulated. The control of the dynamic self-assembly process toward the desired final configuration will be studied through the multiresolution approach, described in the section on designing the energy landscape.

Phase 1. The general strategy for selecting the positions and intensities of external charges to ensure a high probability of having 5 nanoparticles in the top half and 3 nanoparticles in the bottom half of the domain was described in the section on designing the energy landscape. As shown in Figure 8, the first phase is divided into five steps, where the positions and strengths of the external charges change at every step. In the zeroth step, a strong attractive charge is placed in the center

DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384

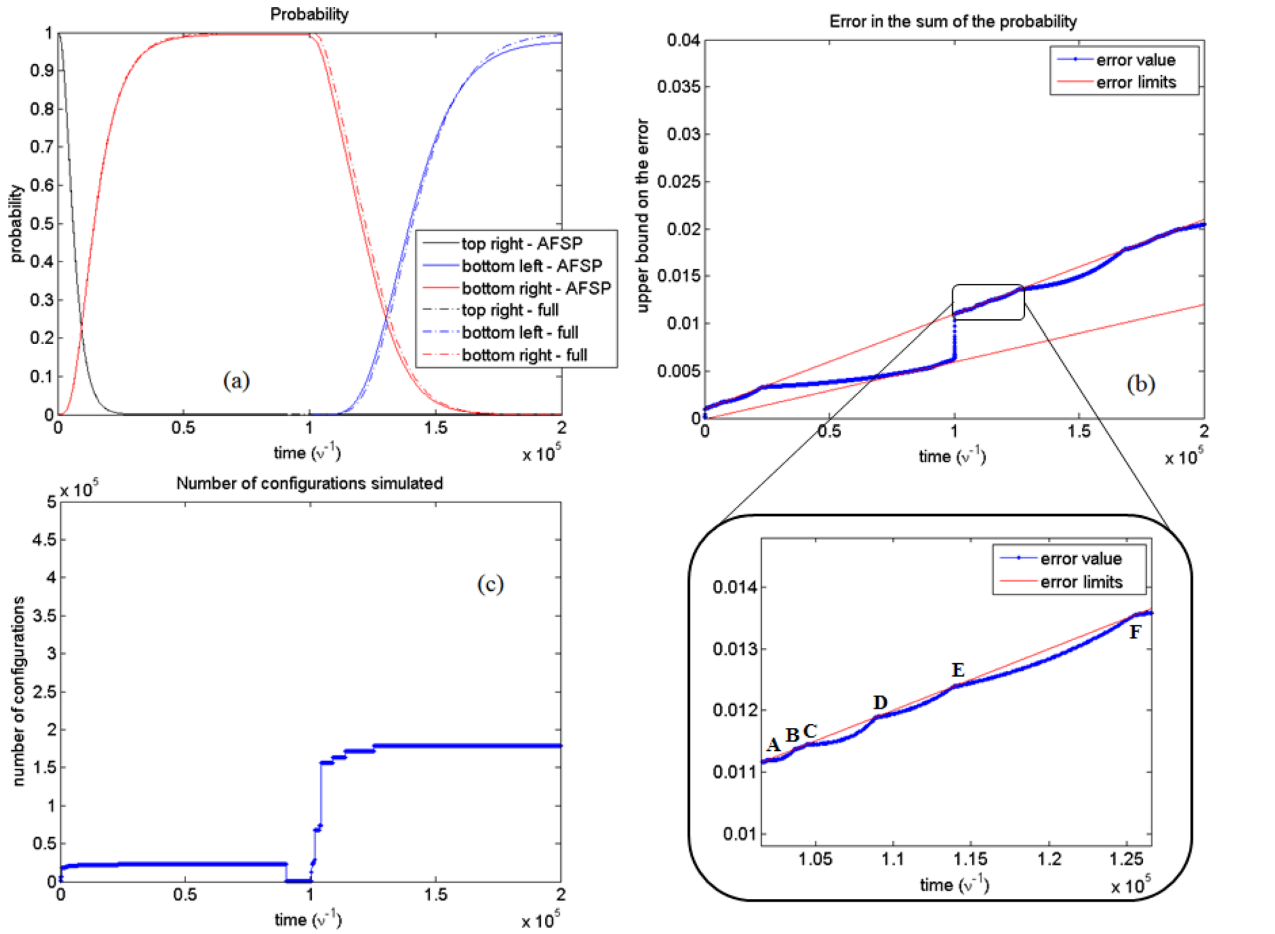

Figure 7. Graphs showing the performance of the proposed method for a case study. The probability of finding nanoparticles in various parts of the domain as a function of time is shown in panel a. The upper bound on the error resulting from model reduction is shown in panel b, where initially events are detected due to the upper bound on the error hitting the maximum value allowed. Reaching steady state results in event detection on the minimum value allowed. After sometime, the upper bound on the error increases rapidly due to the change in the arrangement of external charges and the same procedure as before is repeated. The size of the projection space is shown in panel c, which shows that around 1% of the total configurations need to be simulated.

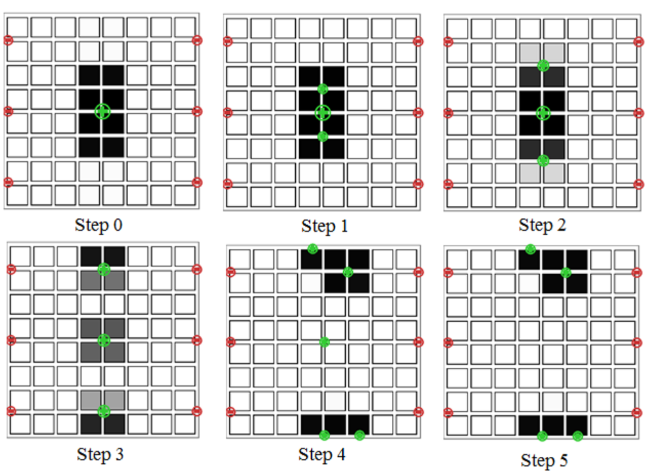

Figure 8. Figure illustrating the location of the external charges in the various steps for Phase 1 of the multiresolution approach. The locations of nanoparticles at the end of each step are also shown in the figure. The gray scale of each cell indicates the expected number of particles that will occupy that cell, which varies from 0 (white) to 1 (black).

of the grid, causing a deep potential well attracting all the nanoparticles. The simulation starts at the end of the zeroth step. The subsequent steps follow the proposed strategy, described in the section on designing the energy landscape: two point charges are introduced on either side of the center charge; new charges move away from the center and their intensities increase; the center charges intensity decreases with time. In the final step, the external charges in the corner are decomposed into two in order to implement the static solution,[21] which stabilizes the steady state configurations and maximizes the probability of the desired arrangement of nanoparticles. Table 2 contains the strengths of the attractive external charges used during each stage.

Table 2. Parameter Values for the Simulation of Phase 1 of the Multiresolution Approach[a]

step	strength of attractive external charges (q_k/q_p) (clockwise)
0	200
1	24, 160, 20
2	48, 120, 40
3	72, 80, 60
4	18.2, 100, 40, 86.5, 39.9
5	18.2, 100, 86.5, 39.9

[a]The strengths of the repulsive charges used are −300 in all the steps. Other parameters used in the simulation are $k_C q_p^2$ = 1 kcal mol^{-1} nm, $k_B T$ = 0.7 kcal mol^{-1}, and a = 10^5 kcal mol^{-1} nm^6.

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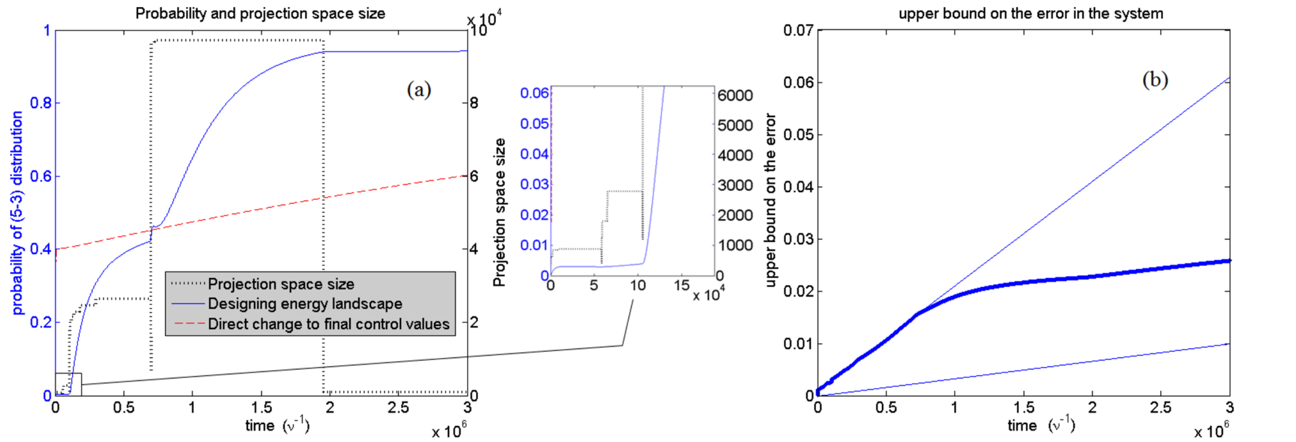

Figure 9. Graphs showing the performance of the proposed strategy for self-assembly in Phase 1 of the multiresolution approach. (a) The blue solid line shows the probability of observing five nanoparticles in the top half and three nanoparticles in the bottom half of the domain. The red dotted line is the probability of the system if we used the control structure of step 5 at time $t = 0$, indicating that a time-dependent strategy is essential for reaching the desired decomposition with high probability. The black dotted line shows the number of configurations simulated (projection space size) by the AFSP at every instant (secondary axis). (b) Graph of the upper bound on the error of the system.

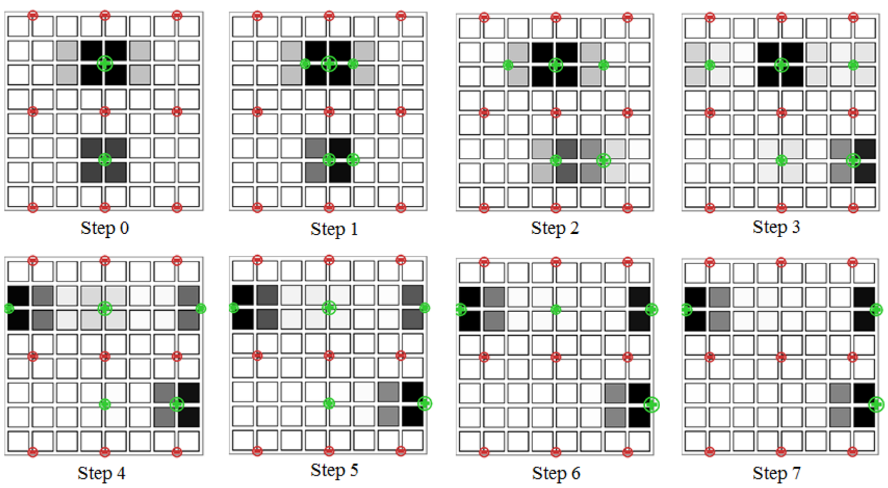

Figure 10. Figure illustrating the location of the external charges in the various steps for Phase 2 of the multiresolution approach. The locations of nanoparticles at the end of each step are also shown in the figure. A horizontal line of repulsive charges is added along the middle of the domain to separate the top and bottom halves of the domain, and thus the two parts are simulated independently.

The probability of observing the system in the desired multiresolution phase from the simulations is shown in Figure 9. The simulation starts with the initial configuration of particles as shown in step 0 of Figure 8 and the structure of external charges as shown in Step 1. The probability of the desired arrangement of nanoparticles (five in the top domain and three in the bottom domain) increases slowly during the simulation of Steps 1 and 2, and hence, the system is quickly moved to Step 3 of the control strategy. This is shown as a magnification in the same figure. In Step 3, a more rapid increase in probability is observed. The simulation at each step is continued until the rate of increase in probability becomes low, after which point the structure of external charges is changed and all the unimportant configurations from the projection space are removed. Note that the removal of unimportant configurations occurs each time the structure of external charges changes. This is in addition to the case when the upper bound on the error of the system hits the minimum allowed value. The probability of the desired arrangement of nanoparticles is shown by a blue line in the graphs and the number of configurations simulated is shown by the black dotted line. The discrete steps in the size of the projection space correspond to the various steps, that is, the changes in charge structure. Figure 9b contains the upper bound on the error of the system that results from the use of the adaptive finite state projection method while performing the simulation, which stays within its limits.

To demonstrate the main advantage of this proposed control strategy, an additional simulation was done, in which the external charges were directly changed to those of Step 5 from Step 0. The probability of the desired arrangement of nanoparticles for that case is plotted as a red dotted line in Figure 9a. As seen in the graph, this procedure yields a self-assembly process with much slower dynamics, and underlines the strength of the proposed control strategy to simulate the dynamics of directed self-assembly in an efficient way.

Phase 2. The aim of this phase is to achieve, with a high probability, an arrangement of three nanoparticles on the top-left part of the grid, two nanoparticles on the top-right part of the grid, and three nanoparticles in the bottom-right part of the grid. At the end of the first phase of the multiresolution approach, repulsive charges are placed at the center of the domain, dividing the top and bottom domains (Figure 10).

DOI: 10.1021/ie504182w
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This is done to create an energy barrier to prevent transitions from the top domain to the bottom domain, and thus break the ergodicity of the system over the whole domain. This allows the two domains to be simulated independently. Similarly as for Step 0 of Phase 1, strong attractive charges are added in the center of each domain (top and bottom half) and simulations are carried from the end of the Step 0, when all the nanoparticles have reached the center of the domain. From Step 1 onward, the proposed control strategy is used to facilitate self-assembly. The strengths of the attractive external charges can be found in Table 3. The strengths of the external charges, when they reach the boundary of the domain, are determined by the static solution.[21]

Table 3. Parameter Values for the Simulation Phase 2 of the Multiresolution Approach[a]

step	strength of attractive external charges (q_k/q_p) (clockwise)
0	150, 100
1	24, 120, 10, 20, 80
2	30, 110, 15, 40, 60
3	40, 100, 20, 60, 40
4	120, 30, 57.1, 60, 40
5	120, 30, 57.1, 100, 20
6	120, 30, 57.1, 100
7	120, 57.1, 100

[a]The strengths of the repulsive external charges used are −100 in all the steps. Other parameters used in the simulation are $k_C q_p^2$ = 1 kcal mol^{-1} nm, $k_B T$ = 0.7 kcal mol^{-1}, and a = 10^5 kcal mol^{-1} nm^6.

The probability of observing the system in the desired multiresolution Phase, from the simulation is shown in Figure 11. The blue line shows the probability of the desired arrangement of nanoparticles, and the green vertical lines in Figure 11a are indicative of the various steps in the control strategy. Note that the charges in the top domain from Steps 1 to 4 (Figure 10) move rapidly, because of the slow rate of increase in probability of the desired arrangement of nanoparticles (three and two particles in the top left and right parts). Since this movement of external charges is rapid, the time points of these changes in steps are very close (Figure 11a). The magnified time points of these step changes are shown as an inset in the same figure. These step changes seem farther apart in Figure 11b because of the difference in time scale. The step changes shown in Figure 11a correspond to Steps 2, 3, 4, and 7, respectively, and Steps 2, 3, 5 and 6, respectively, in Figure 11b. As in the previous phase, the red dotted lines in the plot are the probabilities of the desired arrangements of nanoparticles in the top and bottom halves, when the controls of Step 7 are applied after Step 0, an indication of faster self-assembly through the multiresolution control strategy.

The implementation of various steps in the top half of the grid is done by including all the configurations of the top half of the domain (201 376 configurations). Similarly, the bottom half of the domain is assumed to be an independent system and the simulation is done by considering all possible configurations (4960 configurations).

Phase 3. The final desired configuration of the nanoparticles is obtained with high probability through the implementation of the steps shown in Figure 12. At the end of Phase 2, repulsive external charges are added in the middle of the grid to create an energy barrier and restrict the systems ergodicity in their respective parts of the grid. Therefore, this creates four 4 × 4 grids that can be simulated independently. Each of these grids are small enough to be simulated with the full model without having to use the AFSP method. The strengths of the various external charges used in all the steps can be found in Table 4.

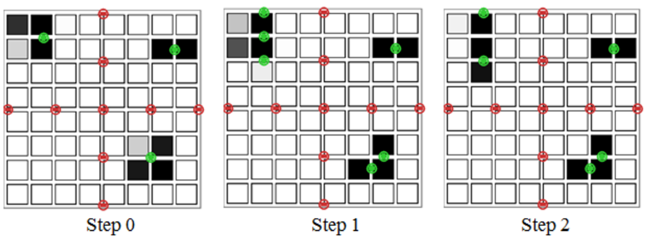

Figure 12. Figure illustrating the location of the external controls in the various steps for Phase 3 of the multiresolution approach. The locations of nanoparticles at the end of each step are also shown in the figure. A vertical line of repulsive charges is now added along the middle of the domain to separate the domain in to four parts, and thus each one is simulated independently.

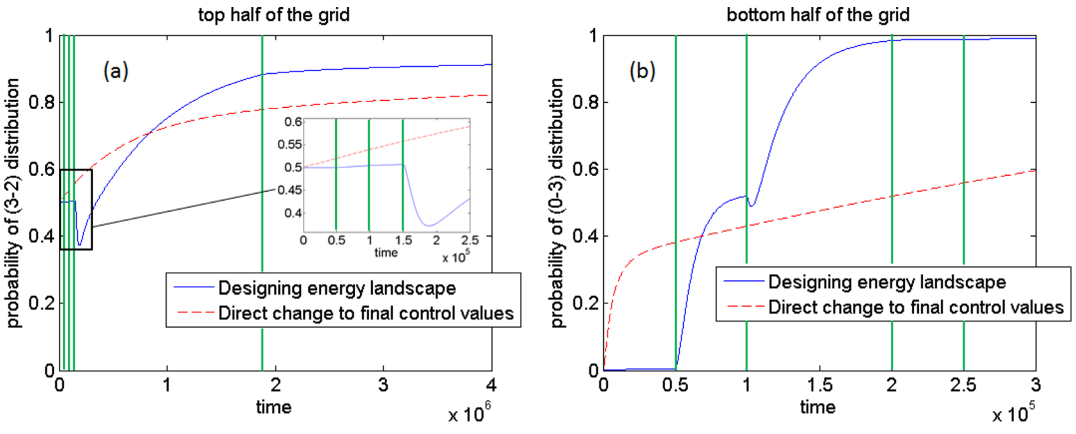

Figure 11. Graphs showing the performance of Phase 2 of the multiresolution approach for the case study. (a) Probabilities of finding three particles and two particles on the left and right sides of the top half of the domain, respectively. The green vertical lines indicate the time points when the external charges switch to the arrangements shown in Steps 2, 3, 4, and 7, respectively. (b) Probability of finding three particles on the bottom-right quadrant of the domain. The green vertical lines indicate the time points when the external charges switch to the arrangements shown in Steps 2, 3, 5, and 6, respectively.

DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384

Table 4. Parameter Values for the Simulation of Phase 3 of the Multiresolution Approach[a]

step	strength of attractive external charges (q_k/q_p) (clockwise)
0	150, 150, 150
1	45, 40, 100, 150, 75, 75
2	45, 100, 150, 75, 75

[a]The strengths of the repulsive external charges used are −50 in all the steps. Other parameters used in the simulation are $k_C q_p^2 = 1$ kcal mol^{-1} nm, $k_B T = 0.7$ kcal mol^{-1}, and $a = 10^5$ kcal mol^{-1} nm^6.

The probability of observing the system in the desired multiresolution phase from the simulations is shown in Figure 13. The probability indicated in the graphs are that of the desired configuration of the nanoparticles in each quadrant. The simulation is started from the end of Step 0, where there is an attractive charge in each quadrant of the system. In Step 1, two attractive charges are added to the top-left quadrant and two attractive charges replace the one attractive charge in the bottom-right quadrant. These locations of the attractive external charges are determined based on the positions of the final external charges (Step 2 of Figure 12). This final external charge structure can be determined using the minimum tiling approach proposed by Solis et al.[21] As the simulation begins, the system quickly changes to Step 2 of the arrangement of charges, due to the slow rate of increase in the probability of the desired configuration in the top-left quadrant. The green vertical lines in Figure 13a indicate the times when the system changes to Step 2.

■ CONCLUSIONS

An algorithm is presented to simulate the dynamics of directed self-assembly of nanoparticles, through the model reduction of master equations. The approach is based on a novel adaptive version of the finite state projection method. Event detection is used to determine effectively those points in time at which the projection space should be modified. Consequently, the method does not require a priori information for changes in projection space, which is particularly useful when the relevant time scales for modification of the projection space vary significantly, as is typically the case for directed self-assembly. Furthermore, transition rates are analyzed systematically to anticipate the configurations that will become relevant to prevent a steep increase in the error from model reduction. A case study illustrates the effectiveness of the approach. Frequent changes in projection space and the number of selected master equations is at least 100-fold smaller for the studied cases, compared to the simulation of the full system.

A control strategy that allows efficient implementation of AFSP is also presented that provides an effective path toward reliable self-assembly. The strategy is based on a multiresolution view of the self-assembled system. Over time, it judiciously decomposes the state space into nonergodic subspaces, by creating new and shifting energy wells, thus moving the nanoparticles along with them. A case study illustrates the proposed strategy and demonstrates the advantage over static strategies, which would have attempted to reach the final desired structure through implementation of the static solution at the initial time. The strategy enables the nanoparticles to follow a dynamic path toward self-assembly that avoids any kinetic traps. The resulting control policy is not optimal in any sense. The strengths of the charges employed were determined *a priori* by trial and error. In the upcoming paper, Part 4 of this series, we will present an optimal control strategy, where an optimum time profile for the locations and strengths of the external charges will be calculated. This will overcome the lack of optimality used in this case study, and allow the fabrication of the desired final structure in minimum time.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors gratefully acknowledge support of this work by the National Science Foundation Grant No. CBET-1033533.

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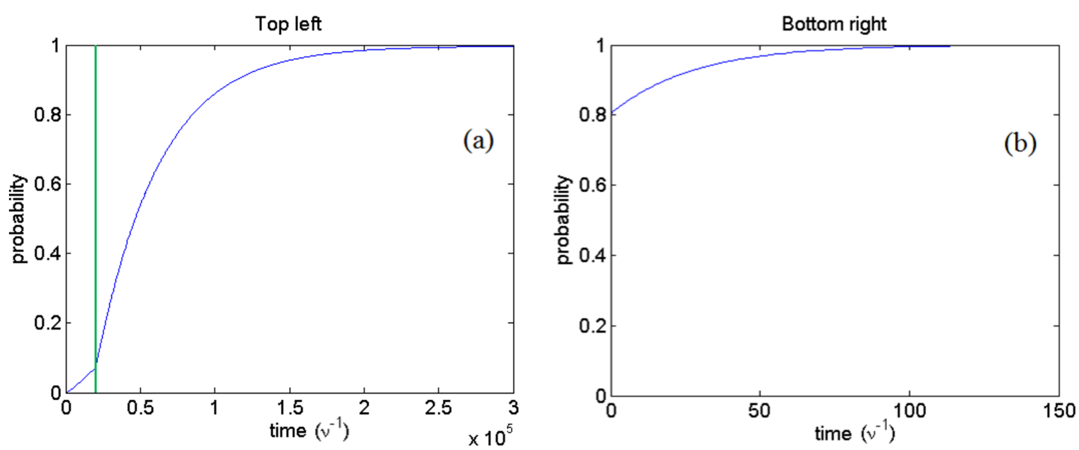

Figure 13. Graphs showing the performance of Phase 3 of the multiresolution approach for the case study. Panels a and b show the probability of the desired configuration of nanoparticles in the top-left and bottom-right quadrants of the domain. This desired configuration is shown in Step 2 of this phase. The green vertical line in panel a indicates the time the external charges switch to the arrangement shown in Step 2.

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DOI: 10.1021/ie504182w
Ind. Eng. Chem. Res. 2015, 54, 4371−4384